

ENTERPRISE HEAVY EQUIPMENT & OPERATIONS ERP

Jainex Heavy Fleet & Crew Operations & Logbook ERP

A Next-Generation Digital Platform for Heavy Crane Fleet Management, Dynamic Gang Deployment, Real-Time Field Attendance, and Physical-Standard Daily Crane Logbooks.

CLIENT ORGANIZATION

Jainex Pariwahan Pvt. Ltd.

CORE MODULES

Fleet · Workforce · Logbook · Deployments

TECHNOLOGY STACK

React + Node.js + Express + MySQL

ACCESS ARCHITECTURE

Multi-Role RBAC (Admin, Sup, Op, Rig)

Executive Summary & Operational Vision

Transforming heavy industrial crane rental and logistics operations from error-prone paper workflows to real-time digital intelligence.



Legacy Challenges

PRIOR BOTTLENECKS

- ✓ **Paper Logbook Delay:** Physical handwritten logbooks took weeks to reach head office for billing.
- ✓ **Disjointed Crew Tracking:** Lack of real-time visibility into which Operator and Rigger are assigned to which Crane and Site.
- ✓ **Disputed Engine Hours:** Meter reading mismatches, leading to client billing disputes.
- ✓ **Manual Attendance:** Inability to correlate on-site attendance with crane operation logs.



The Unified Solution

DIGITAL TRANSFORMATION

- ✓ **1-to-1 Digital Logbook:** Pixel-perfect reproduction of physical Jainex logbook sheets with automated hour calculations.
- ✓ **Dynamic Gang Pairing:** Group Cranes + Operators + Riggers into certified operational units.
- ✓ **Role-Tailored Panels:** Customized views for Admins, Supervisors, Operators (OP-xxx), and Riggers (RIG-xxx).
- ✓ **Instant Billing Reconciliation:** Export and print monthly verified client timesheets in 1 click.



Strategic Value

BUSINESS ROI

- ✓ **100% Billing Accuracy:** Elimination of billing discrepancies through validated meter logs.
- ✓ **Zero Data Lag:** Instant daily submission from job sites directly to ERP database.
- ✓ **Asset Protection:** Continuous tracking of crane operational health, KMH, and engine hours.
- ✓ **Strict Role Boundaries:** Tailored mobile-ready portals preventing unauthorized data alteration.

End-to-End Technology Stack

Engineered with high performance, secure micro-modular design and sub-second response times.

FRONTEND CLIENT

React 18 + Vite

Modern SPA with CSS Design Tokens

React Router v6Context APIAxios Interceptors

Lucide IconsPrint Media CSS

BACKEND API

Node.js + Express

RESTful Micro-Services Controller Pattern

Express.jsJWT Auth TokensRole Guards

Multer UploadsCORS Middleware

DATABASE ENGINE

MySQL 8.0 Enterprise

Relational Normalized Schema with Foreign Keys

InnoDB EngineConnection PoolACID Compliant

Date/Hour IndexingCascade Integrity

SECURITY & STORAGE

Security & Assets

Encrypted Credentials & Media Storage

BCrypt HashSanitized Payloads

Static Media CDNID Prefix Routing

SSL / TLS Ready

100%

RESTful Endpoint Coverage with JWT Verification

< 40ms

Average MySQL Query Latency for Monthly Logbook Aggregations

Responsive

Seamless UI for Heavy Crane Field Tablets, Mobile Phones & Desktops

Multi-Tier Role-Based Access Control (RBAC)

Strict separation of duties between Administrative staff, Field Supervisors, Crane Operators, and Rigging Crew.

👑 Admin

FULL ACCESS

Executive & Central Operations authority across the entire company fleet.

✓ Create & manage Cranes, Operators, Riggers & Sites.

✓ Assign & transfer Gangs across client sites.

✓ View, edit, approve & print all crane logbooks.

✓ Access comprehensive audit & financial stats.

🛡️ Supervisor

FIELD LEAD

On-site operations management and workforce compliance.

✓ Deploy gangs & manage daily site resource allocations.

✓ Audit workforce daily attendance logs.

✓ Monitor crane operational hours & status.

✓ Review and verify submitted operator log sheets.

🚧 Operator

OP-XXX ID

Licensed Crane Operators driving daily machine operations on-site.

✓ Personalized operator dashboard & profile.

✓ Self attendance check-in & shift logging.

✓ **Daily Crane Logbook Entry:** Record hours, KMH, party, fuel & downtime.

✓ Autofill current crane and active site details.

👷 Rigger

RIG-XXX ID

Heavy rigging specialists assisting crane lifting and load safety.

✓ Personalized rigger dashboard & profile.

✓ Daily crew attendance logging.

✓ View assigned crane, partner operator & active site.

✓ **No Logbook Access:** Focused solely on crew duty & site operations.

Role Boundary Enforcement: Dual-layer verification in place — frontend `ProtectedRoute` guards prevent route access, while backend Express middleware (`authorizeRoles`) blocks unauthorized REST API requests at the database layer.

Fleet & Asset Master Management

Centralized registry of heavy crane machinery, technical specifications, operational status, and document compliance.



Comprehensive Crane Attributes

TECHNICAL MASTER DATA

Every crane in the fleet is tracked with high-precision specifications to prevent equipment mismatch during client deployments.

Identifiers

Reg No, Chassis No, Engine No, Model

Lifting Capacity

25 Ton – 500+ Ton Hydraulic & Crawler

Boom Length

Main Boom + Fly Jib Extensions

Ownership Type

Owned vs Hired / Leased

- ✓ Photo gallery uploads via encrypted file pipeline.
- ✓ Compliance expiry tracking (Fitness, Insurance, Third-Party Safety Certs).



Live Asset States & Lifecycle

STATE MACHINE

STATUS	OPERATIONAL MEANING	SYSTEM BEHAVIOR
ACTIVE	Deployed on active client project site	Available for daily logbook entries & gang operations
IDLE / YARD	Available in base yard for fresh deployment	Open for assignment to new client site contracts
MAINTENANCE	Under active mechanical or electrical overhaul	Locked from new deployments until certified fit

Auditable History: All status transitions and site reassignments are permanently logged with timestamps to maintain complete lifecycle traceability.

Operators & Riggers Profile Architecture

Digital personnel management for certified heavy machinery operators and specialized rigging crew.



Crane Operators (Prefix: OP-xxx)

MACHINERY CAPTAINS

Operators are assigned dedicated ID codes (e.g. OP-001, OP-002) which drive URL routing, login authentication, and logbook entries.

- ✓ **Heavy Vehicle License:** License number, category & expiration tracking.
- ✓ **Operating Experience:** Certified experience on Hydraulic, Crawler & All-Terrain cranes.
- ✓ **Photo & Emergency Contacts:** Full KYC profile with Aadhaar/Govt ID verification.
- ✓ **Logbook Authorization:** Granted sole permission to input daily engine meter readings.



Riggers & Signalmen (Prefix: RIG-xxx)

SAFETY & RIGGING CREW

Riggers are assigned specialized ID codes (e.g. RIG-001) with dedicated dashboards focused on attendance and site deployment.

- ✓ **Rigging Certification:** Heavy lift slings, shackles, and signal certification records.
- ✓ **Gang Pairing Linkage:** Automatically connected to their assigned Crane and Operator.
- ✓ **Streamlined Experience:** Distraction-free interface with logbook module excluded for focused daily site attendance.
- ✓ **Real-Time Assignment View:** Live view of assigned client project and active supervisor.

OP / RIG

Automated Sequential Zero-Padded ID Generation

Instant KYC

Document Uploads & Verification directly to Server Storage

1-Click Pair

Instant Assignment to Heavy Crane Gang Units

Site Master & Dynamic Gang Deployment

Coordinating complex heavy equipment logistics by pairing certified human capital with heavy machinery.



Site Master

PROJECT GEOGRAPHIES

Comprehensive tracking of client construction yards, refineries, bridge sites, and industrial complexes.

- ✓ Client / Party Name (e.g. L&T, Vaksim Construction).
- ✓ Site Location, State, GPS Landmark & In-charge Contact.
- ✓ Contract commencement & estimated demobilization dates.
- ✓ Active resource count currently stationed on-site.



The "Gang" Operational Concept

TRIAD RESOURCE UNIT



Crane
Primary Asset



Operator
Licensed Driver



Rigger
Safety Specialist



Project Site
Deployment

Seamless Resource Transfers
Transfer Cranes or Crew between active sites with automatic assignment history archiving.

Release & De-Mob Management
Release units back to pool immediately upon job completion with 1-click status resets.

Real-Time Field Attendance System

Capturing daily field presence, shift schedules, overtime hours, and attendance verification.



Operator & Rigger Self-Mark

FIELD MOBILE UI

Crew members can mark their daily attendance directly from their mobile devices with clean, large-button interfaces.

- ✓ Automated timestamp and date capture.
- ✓ Visual calendar view of past attendance history.
- ✓ Status selection: Present, Absent, Half-Day, Sick Leave.



Supervisor Site Roster

BULK VERIFICATION

Site Supervisors can view and verify full-crew presence across all cranes operating on their assigned project.

- ✓ Bulk attendance marking for entire rigging gangs.
- ✓ Immediate red-flag alerts for missing personnel.
- ✓ Site-level daily muster roll export for client safety audits.



Admin Payroll Integration

HR & ACCOUNTING

Central HR and Payroll teams receive verified monthly attendance aggregates for payroll disbursement.

- ✓ Monthly aggregate sheets filtered by role & employee.
- ✓ Overtime hours correlation with crane logbook records.
- ✓ Zero ghost-worker claims through audited database logs.

Cross-Module Synchronization: Operator attendance records are cross-checked with crane daily logbook entries to guarantee that machinery was operated only by verified on-duty staff.

Crane Daily Digital Logbook

Direct 1-to-1 digital replica of the physical handwritten Jainex Pariwahan Pvt. Ltd. log sheet.



Authentic 14-Column Physical Sheet Replica

PHYSICAL-STANDARD DESIGN

Designed specifically to match the exact physical paper format used across India's heavy infrastructure sites for decades:

COL #	HEADER FIELD	OPERATIONAL FUNCTION
01-03	Date · Shift · Crane Reg No	Operational date, day/night shift & asset tag
04-06	Client Party · Site · Job Description	Work order client & specific lifting task
07-09	Start Time · Finish Time · Total Hours	Shift operational duration & billing units
10-12	Engine Hour Meter (Initial/Final/Run)	Exact machine run time for diesel/maintenance
13-14	KMH Run · Breakdown / Idle Hours	Transit distance & contract downtime logging



1-Click Fast Entry

FIELD ERGONOMICS

- ✓ **Autofill Active Deployment:** When an Operator opens the form, their assigned Crane, Site, and Party are auto-populated.
- ✓ **Auto-Calculation:** Total working hours, engine runtime, and KM differences calculate automatically as digits are entered.
- ✓ **Instant Validation:** Prevents invalid inputs (e.g. final meter lower than initial meter).
- ✓ **Zero Learning Curve:** Familiar form layout empowers operators with minimal technical background.

Monthly Log Sheet Verification, Print & Export

Automating monthly client invoice verification, print-ready formal log sheets, and spreadsheet exports.



Print-Ready Official Sheet

CLIENT SUBMISSION

Dedicated CSS print media styling turns digital logbook data into an authentic Jainex Pariwahan Pvt. Ltd. printed sheet with complete legal headers and signatures.

- ✓ Company header & registered office address.
- ✓ Party name, site name, and crane reg number.
- ✓ Dedicated signature blocks for Operator, Jainex Site In-Charge, and Client Representative.



Real-Time Summary Ribbon

DYNAMIC KPI BAR

Instant statistical ribbons dynamically calculate KPIs for any selected Crane and Month:

Total Hrs Net Billable Shift Duration	Engine Run Hour Meter Difference
Total KMH Odometer Distance Run	Breakdowns Total Downtime Hours



1-Click CSV / Excel Export

DATA PORTABILITY

Export filtered monthly logs directly into standardized CSV formats for seamless ingestion into ERP / Tally accounting packages.

- ✓ Auto-formatted column headers.
- ✓ Direct integration with corporate finance audits.
- ✓ Eliminates duplicate manual data entry for accountants.

Relational Database Schema & Data Integrity

Normalized MySQL database architecture ensuring relational consistency and transaction safety across all operations.



Core Relational Entities

TABLE ARCHITECTURE

TABLE NAME	PRIMARY KEY / IDENTIFIER	CORE RELATIONSHIPS & ROLE
cranes	id (INT AUTO)	Fleet master data, capacity, model, status
operators	id (INT AUTO)	Operator profile, license, link to users
riggers	id (INT AUTO)	Rigger profile, experience, link to users
sites	id (INT AUTO)	Project location, party, supervisor link
gang_assignments	id (INT AUTO)	Pairs crane_id, operator_id, rigger_id
deployments	id (INT AUTO)	Links Gang or Crane to site_id with dates
logbook_entries	id (INT AUTO)	Daily log sheet records, meter readings, hours



Data Validation & Consistency Rules

ACID RULES

- ✓ **Referential Integrity:** Foreign key constraints guarantee that logbook entries cannot be created without valid Crane and Site records.
- ✓ **Unique Constraint Protection:** Prevents duplicate attendance or conflicting gang assignments for the same operator on a single date.
- ✓ **Audit Trail Fields:** Every record tracks created_at, updated_at, and originating user_id.
- ✓ **Connection Pooling:** High-throughput MySQL connection pool manages concurrency under peak field check-in hours.

Security: All SQL queries utilize parameterized statements to completely prevent SQL injection attacks.

Business Impact & Future Strategic Roadmap

Delivering measurable operational efficiency today while laying the foundation for smart telematics tomorrow.



Immediate Business Gains

DELIVERED VALUE

- ✓ **90% Reduction** in invoice verification turnaround times.
- ✓ **Zero Log Loss:** 100% digital redundancy of all heavy machinery operating logs.
- ✓ **Clear Accountability:** Pinpoint exact gang responsibility for crane health and fuel usage.
- ✓ **Total Field Visibility:** Real-time overview of all machine deployments across all client sites.



Enterprise Security Standards

SECURITY BASELINE

- ✓ **JWT Authentication:** Secure encrypted session tokens.
- ✓ **Role-Guarded Endpoints:** Automatic blocking of unauthorized privilege escalations.
- ✓ **Sanitized Media Uploads:** Safe image processing for fleet and crew documents.
- ✓ **Daily Automated Backups:** Resilient disaster recovery procedures for all database records.



Future Phase 2 Roadmap

FUTURE HORIZONS

- ✓ **IoT Telematics Integration:** Direct CAN-bus integration to automatically read engine hours and fuel meters.
- ✓ **GPS Geofencing:** Automatic site check-in when crew/crane enters client yard boundary.
- ✓ **Predictive Maintenance:** Automated servicing reminders based on accumulated engine hours.
- ✓ **Offline-First Mobile App:** Service worker support for remote sites without network coverage.

Thank You!

Jainex Heavy Fleet & Crew Management System · Built for Enterprise Operational Excellence